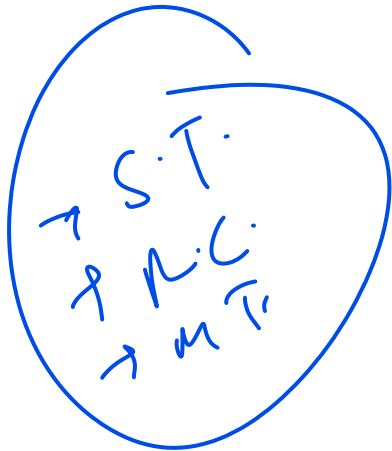


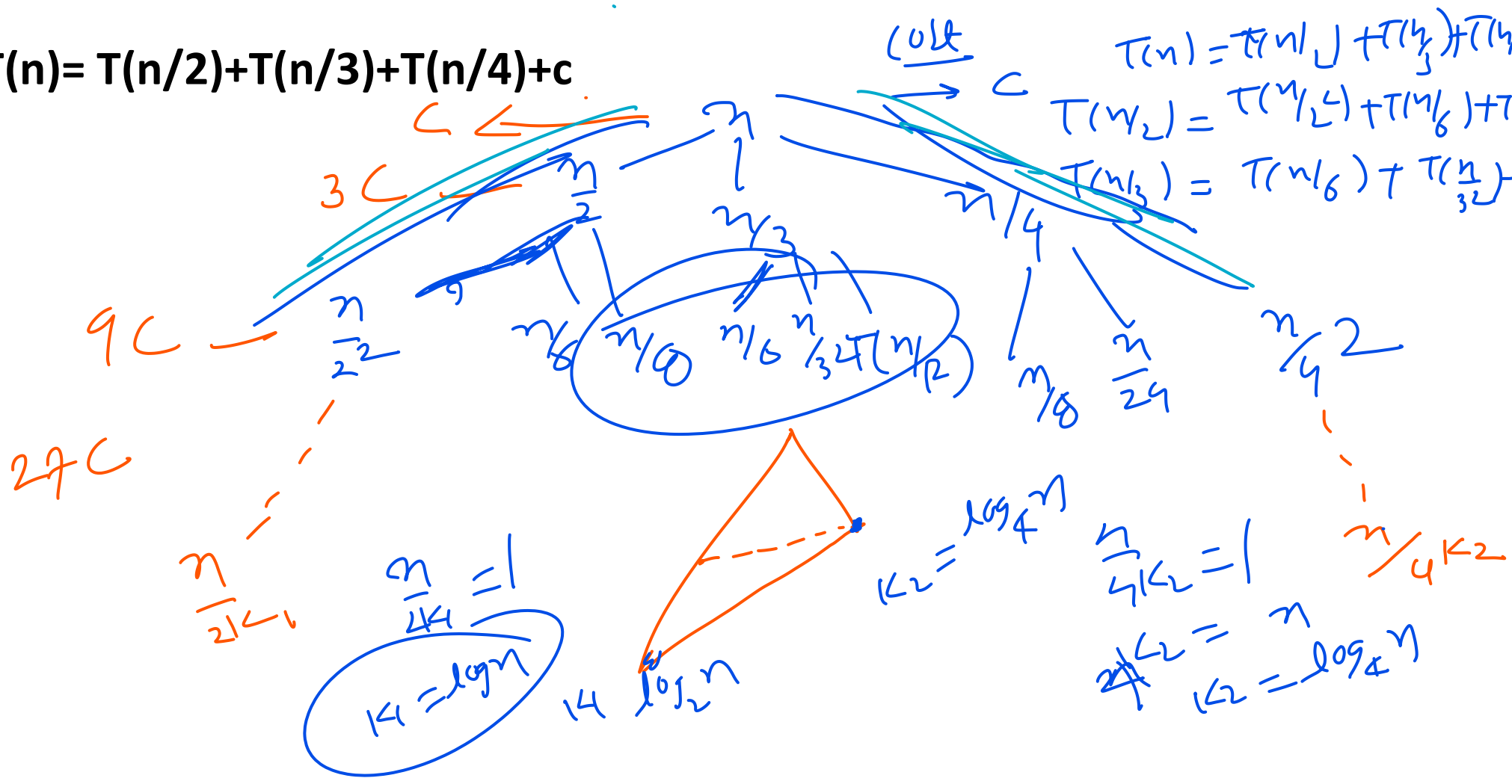
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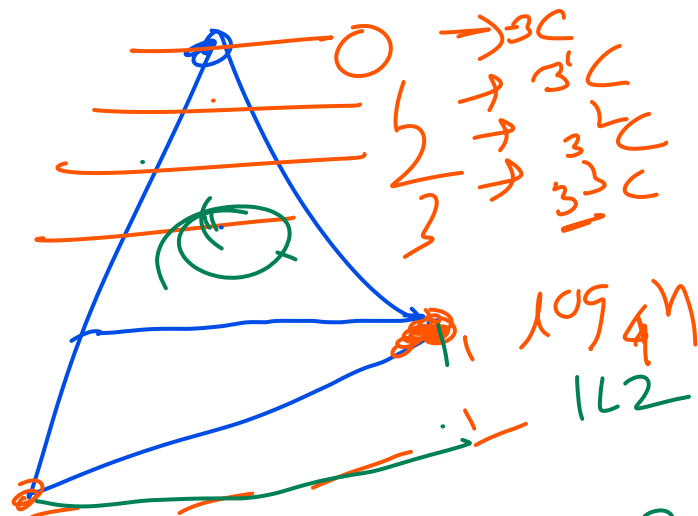
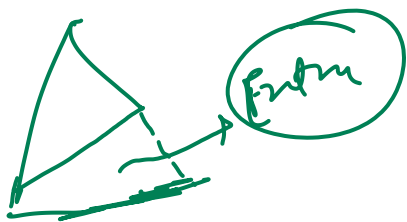
Recursive Tree Method



marker function

5.) $T(n) = T(n/2) + T(n/3) + T(n/4) + c$





$$|C| = \log_2 n$$

$$T(n) \leq 3^0 C + 3^1 C + 3^2 C + \dots + 3^{|C|} C$$

$$T(n) \leq 3^{|C|} C$$

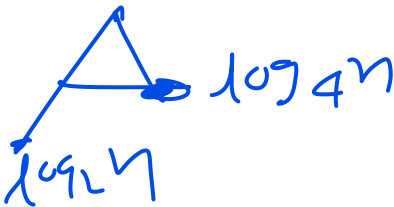
$$T(n) \leq \frac{3^{n+1} - 3}{3 - 1}$$

$$T(n) \leq \frac{3^{n+1} - 3}{2}$$

$$T(n) \leq 3^{n+1}$$

$$T(n) = O(3^n)$$

$$n = \log_2 n \Rightarrow O(n \log_2 n)$$



$$T(n) \geq 3^0 + 3^1 + 3^2 + \dots + 3^{k_2} c$$

$$T(n) \geq c(3^0 + 3^1 + 3^2 + \dots + 3^{k_2})$$

$$= c \frac{3^{k_2+1} - 1}{3 - 1}$$

$$T(n) \geq c(3^{\log_4 n})$$

$$\geq c(n^{\log_4 3})$$

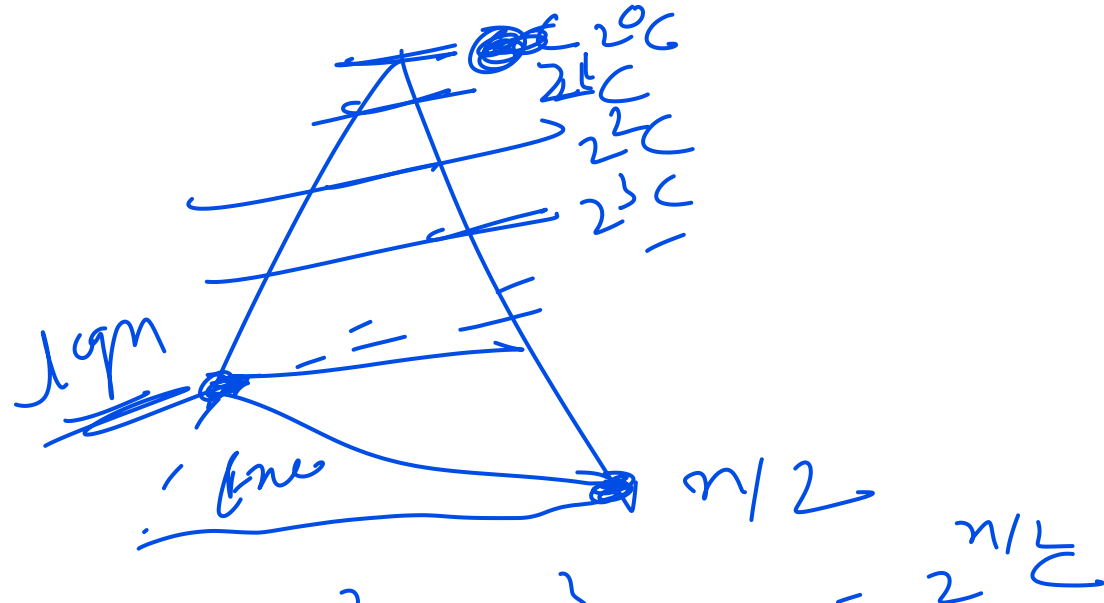
$$T(n) \leq 3^{k_1}$$

$$T(n) \leq 3^{k_2}$$

$$k_2 = \log_4 n$$

$$T(n) = \Theta(3^{k_1})$$

$$T(n) = \Omega(3^{k_1}) \quad k_1 = \log_2 n$$



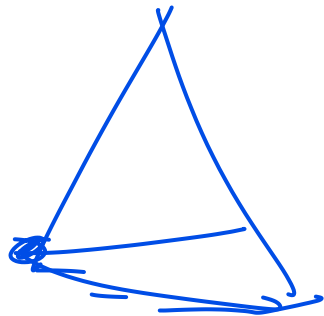
$$T(n) = 2^0 C + 2^1 C + 2^2 C + 2^3 C + \dots + 2^{n/2} C$$

$$= C \left[\frac{a(r^n - 1)}{r - 1} \right]$$

$$= C \left[\frac{1(2^{n/2} - 1)}{2 - 1} \right]$$

$$T(n) \leq \frac{2^{n/2} - 1}{2 - 1}$$

$$T(n) = O(2^{n/2})$$



$$T(n) \geq 2^0 c + 2^1 c + 2^2 c - \dots - \frac{2^{\log_2 n} c}{2}$$

$$= c \left(1 + 2 + 2^2 + \dots + 2^{\log_2 n - 1} \right)$$

$$\geq c 2^{\log_2 n - 1}$$

$$T(n) \geq n^{\log_2 2}$$

$$T(n) = \Omega(n)$$

$$T(n) = n$$

$$T(n) \leq 2^{n/2}$$

$$T(n) \geq n$$



Thanks

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